Original Article

Assessment of Stability of Fluoride Vials Over a Time Period

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Abstract
Background: Blood glucose estimation is one of the commonest and most widespread tests in any clinical laboratory. Rising concern about diabetes mellitus and metabolic syndrome emphasizes correct evaluation of blood glucose. Grey-top fluoride vials are used in most laboratories, though their stability over time is not well characterized. **Objectives:** To predict the stability of grey-top sodium fluoride (NaF) vials, so as to minimize erroneous glucose results when laboratory turnaround time is increased or repeat measurement from the same vial is required. **Methods:** Twelve retained blood glucose samples (across four clinically relevant glucose levels) were rerun at six time points — baseline, 24, 48, and 72 hours, and 7 and 14 days after collection — on a Vitros 4600 dry chemistry system using the glucose oxidase-peroxidase method. **Results:** A statistically significant difference from baseline was observed at 24 hours (paired t-test, $p=0.005$), whereas differences at 48 hours, 72 hours, 7 days, and 14 days were not statistically significant ($p>0.05$). Mean $\pm$ SD glucose values were 155.50 ± 65.13 mg/dL (24 h), 153.33 ± 63.49 mg/dL (48 h), 153.42 ± 63.65 mg/dL (72 h), 153.42 ± 63.43 mg/dL (7 days), and 154.25 ± 63.60 mg/dL (14 days), against a baseline of 153.75 ± 64.25 mg/dL. **Conclusion:** A statistically significant difference from baseline was detected at 24 hours, but the absolute mean difference was small (1.75 mg/dL). No statistically significant differences were detected at 48 hours, 72 hours, 7 days, or 14 days. These findings suggest that, under the storage conditions used in this study, the observed statistical change at 24 hours should be interpreted cautiously and should not by itself establish a universal 24-hour stability limit for all grey-top fluoride specimens.

Key Words: Glucose, Grey-top vials, Sodium fluoride, Stability

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I. Introduction

Diabetes mellitus remains a rising health concern in developing and developed countries alike. It is a metabolic disease characterized by increased blood glucose levels with absolute or relative insulin deficiency or insulin resistance, and related complications. Community studies indicate that the growing diabetic population is creating an increasing burden of ill health on society[1]. With significant lifestyle changes, healthcare professionals remain concerned about the accuracy and precision of blood glucose measurement — not only for known diabetics, but also for early diagnosis of pre-diabetics, which may lead to better clinical outcomes.

Assurance of the integrity and reliability of clinical laboratory results is a necessity. Challenges to quality results include preanalytical, analytical, and postanalytical errors, as well as biological variation. According to National Academy of Clinical Biochemistry guidelines, in vitro glycolysis must be prevented to avoid erroneous blood glucose estimation[2]. Blood glucose is usually measured in vials containing sodium fluoride (NaF) and sodium oxalate: fluoride, a competitive inhibitor of enolase, inhibits glycolysis and preserves the true glucose value, while oxalate acts as an anticoagulant. The action of NaF is slow, taking almost 4 hours from collection to show its antiglycolytic effect, and remains stable thereafter for 2–3 days[3,4]. A study of 200 pregnant women assessed for gestational diabetes mellitus concluded that fluoride-citrate vials are a better alternative for diagnosis, investigating fluoride, citrate, and ice as antiglycolytic agents[5] — citrate being an inhibitor of phosphofructokinase in the glycolytic pathway.

Vial stability is a prerequisite in a clinical laboratory setting, both for cross-checking values to rule out preanalytical, analytical, and postanalytical error or normal biological variation, and for accommodating delays in sample processing that may occur in practical hospital settings without altering the result and consequent diagnosis. Since glucose is one of the commonest laboratory parameters, any measurement discrepancy may lead to a wrong diagnosis and a serious downstream outcome. Fluoride-treated specimens are more effective than heparin-treated specimens, as the rate of decrease in glucose concentration after the first 4 hours is much slower in fluoride vials[4]. This study aims to predict the stability of grey-top NaF vials, in order to remove erroneous results in blood glucose estimation even where laboratory turnaround time increases.

Objective

To predict the stability of grey-top NaF vials, so as to remove erroneous results in blood glucose estimation even under conditions of increased turnaround time, or where a repeat measurement from the same vial is required due to machine-related technical error.

II. Materials and Methods

The study was conducted in the Biochemistry Laboratory of JMN Medical College, Chakdaha, as a trial run on retained samples, to predict the stability of grey-top glucose vials over a time period. Four levels of blood glucose value of clinical significance were selected — 90, 120, 150, and 250 mg/dL — with a total of 12 samples taken such that three samples in each group had similar glucose levels. Sampling was by convenience, and samples were selected to include fasting, post-prandial, and random blood glucose specimens. Approximately 2 mL of blood was present in all BD grey-top vials, containing potassium oxalate and sodium fluoride. Haemolysed and clotted samples were excluded.

Samples were rerun at 24 hours, 48 hours, 72 hours, 7 days, and 14 days after collection. Glucose was estimated on a Vitros 4600 Dry Chemistry system using the glucose oxidase-peroxidase (GOD-POD) method. Samples were stored at 2–8°C in the Biochemistry Clinical Laboratory freezer between runs.

III. Results

Glucose values in grey-top vials were assessed at six time points (n=12); only the 24-hour measurement differed significantly from baseline. Mean $\pm$ SD glucose levels were 155.50 $\pm$ 65.13

mg/dL (24 h), 153.33 ± 63.49 mg/dL (48 h), 153.42 ± 63.65 mg/dL (72 h), 153.42 ± 63.43 mg/dL (7 days), and 154.25 ± 63.60 mg/dL (14 days), against a baseline of 153.75 ± 64.25 mg/dL; coefficients of variation were 41.89%, 41.41%, 41.49%, 41.34%, and 41.23% respectively. The mean difference from baseline was 1.75 mg/dL at 24 h (p=0.005), 0.42 mg/dL at 48 h (p=0.480), 0.33 mg/dL at 72 h (p=0.368), 0.33 mg/dL at 7 days (p=0.457), and 0.5 mg/dL at 14 days (p=0.293).

Table 1: *Glucose measurements over time in retained grey-top fluoride vials (n=12)*

Time point	Mean ± SD (mg/dL)	Difference; *p* value
Baseline	153.75 ± 64.25	—
24 h	155.50 ± 65.13	1.75; 0.005
48 h	153.33 ± 63.49	-0.42; 0.480
72 h	153.42 ± 63.65	-0.33; 0.368
7 days	153.42 ± 63.43	-0.33; 0.457
14 days	154.25 ± 63.60	0.50; 0.293

IV. Discussion

This study aimed to estimate the stability of grey-top glucose vials over six time points. Only the 24-hour measurement differed significantly from baseline (p=0.005); the differences at 48 hours, 72 hours, 7 days, and 14 days were not statistically significant. The mean glucose level increased by 1.75 mg/dL at 24 hours, decreased by 0.42 mg/dL at 48 hours and by 0.33 mg/dL at 72 hours and 7 days, and increased by 0.50 mg/dL at 14 days. The observed fluctuations were small and may reflect analytical or preanalytical variation; the study design does not allow the specific cause to be established. Notably, the difference remained similar from the third to the seventh day, suggesting that glucose measurements remained relatively stable across those later time points under the study storage conditions.

The statistically significant 24-hour difference was small (1.75 mg/dL) and is unlikely to be clinically important for glucose interpretation in routine laboratory medicine; such minor variation would not typically change a diagnosis or clinical decision. The different glucose-level groups showed broadly similar patterns of change over time. This finding is consistent with Imtiaz et al. (2019), who reported a mean difference in glucose level of 1.2 mg/dL after 24 hours[6].

Glucose estimation from grey-top vials is therefore preferable within a day of collection, though delayed estimation does not appear to carry major clinical significance.

Limitation

The main limitation of this study is its small sample size. A study with a larger sample size is required to give better insight and provide further clarity for laboratory medicine and improved patient care.

V. Conclusion

Under the storage conditions used in this study, the observed mean glucose values remained relatively stable across the study period, with a statistically significant but small difference at 24 hours. Larger studies are needed before defining a universal stability limit for grey-top fluoride specimens.

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